VECTOR-Drive: Tightly Coupled Vision-Language and Trajectory Expert Routing for End-to-End Autonomous Driving

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***Abstract*—End-to-end autonomous driving requires models to understand traffic scenes, infer driving intent, and generate executable motion plans. Recent vision-language-action (VLA) models inherit semantic priors from large-scale vision-language pretraining, yet still face a coupling trade-off: fully shared backbones preserve multimodal interaction but may entangle language reasoning and trajectory prediction, whereas decoupled reasoning-action pipelines reduce task conflict but weaken semantic-motion coupling. We propose VECTOR-DRIVE, a tightly coupled VLA framework built on Qwen2.5-VL-3B. VECTOR-DRIVE keeps all tokens coupled through shared self-attention and routes feed-forward computation according to token semantics. Vision and language tokens are processed by a Vision-Language Expert to preserve semantic priors, while target-point, ego-state, and noisy action tokens are routed to a Trajectory Expert for motion-specific computation. On the action-token pathway, a flow-matching planner refines noisy action tokens into future waypoints and speed profiles. This design couples semantic reasoning and motion planning within a single multimodal Transformer while separating task-specific FFN computation. On Bench2Drive, VECTOR-DRIVE achieves 88.91 Driving Score and outperforms representative end-to-end and VLA-based baselines. Qualitative results and ablations further validate the benefits of shared attention, semantic-aware expert routing, progressive training, and flow-based action decoding.**

***Index Terms*—End-to-end autonomous driving, vision-language-action model, trajectory planning, expert routing, flow matching.**

I. INTRODUCTION

END-TO-END autonomous driving maps multimodal observations directly to future trajectories, enabling perception, prediction, and planning to be optimized in a unified framework [1]–[6]. Recent methods have advanced this paradigm through trajectory-guided control, scalable planning decoders, adapter-based perception-planning decoupling, vectorized scene representations, and unified transformer architectures [7]–[11]. Despite progress on closed-loop benchmarks such as Bench2Drive [4], [12], long-tail interactions and environmental shifts can still cause accumulated planning errors [13], [14].

Vision–Language–Action (VLA) models address this challenge by exploiting large-scale vision-language pretraining and semantic supervision [13], [15]–[17]. Recent driving VLA systems further introduce language-action alignment, vision-language instructed action generation, and adaptive reasoning to improve semantic grounding and decision robustness [17]–[19]. A common design adapts a pretrained VLM or Large Language Model (LLM) to driving by attaching a trajectory or control head to the shared backbone, as illustrated in Fig. 1(a) [13], [14], [17], [20], [21]. While this preserves multimodal interaction, the same transformer and FFN parameters must support both discrete autoregressive reasoning and continuous geometric prediction. Due to their different inductive biases and optimization behavior, fully shared computation may introduce task interference and degrade motion accuracy [22], [23].

A complementary direction separates high-level reasoning from low-level action generation, as shown in Fig. 1(b) [14], [24]–[26]. In such pipelines, a VLM provides semantic representations, while a separate action module or trajectory expert predicts chunk-level motion [15], [24]. Although this reduces optimization pressure on the shared backbone, trajectory generation often depends on semantics through a coarse conditioning interface [17], [27]. This can limit interactive driving, where semantic constraints must be transferred to fine-grained, dynamically feasible trajectories.

We propose **VECTOR-DRIVE** , a tightly coupled VLA framework Fig. 1(c) for end-to-end autonomous driving. Instead of fully sharing or fully separating reasoning and action generation, VECTOR-DRIVE shares self-attention across all tokens and routes FFN computation according to token semantics. Vision and language tokens are processed by a Vision-Language Expert, while target-point, ego-state, and noisy action tokens are routed to a Trajectory Expert. On the action-token pathway, a flow-matching planner transforms noisy action representations into future waypoints and speed profiles. Thus, semantic reasoning and trajectory generation remain coupled through attention, while language-oriented and motion-oriented computation is separated in FFN layers.

We evaluate **VECTOR-DRIVE** on Bench2Drive in CARLA. It achieves 88.91 Driving Score and 71.82% Success Rate. Ablations further isolate the effects of shared attention, semantic-aware FFN routing, progressive training, and flow-based trajectory decoding.

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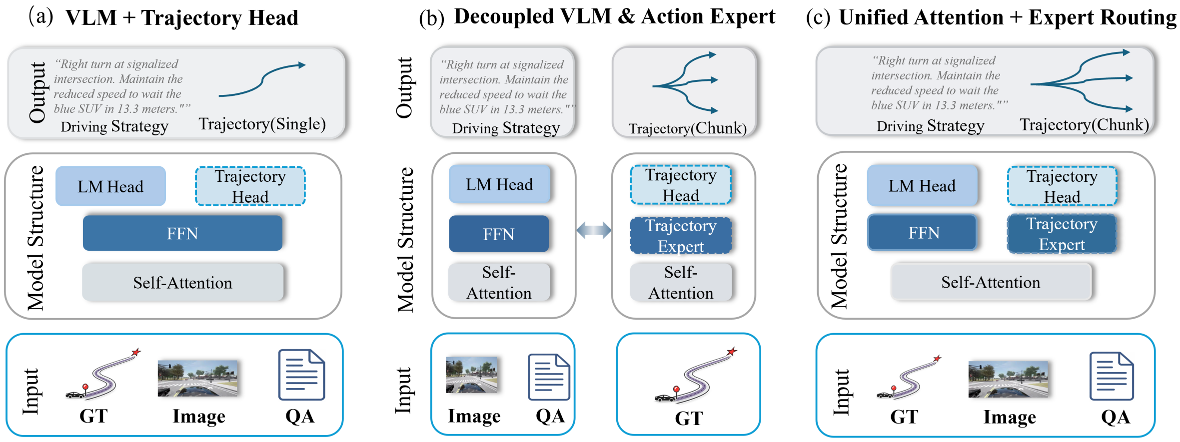

Fig. 1. **Three VLA design paradigms.** Left: a shared VLM predicts actions with a single trajectory head. Middle: reasoning and chunk-level motion generation are separated. Right: our shared-attention and expert-routed design preserves multimodal interaction while routing motion-related computation to a dedicated Trajectory Expert.

The main contributions are:

- We propose a unified VLA architecture that combines **shared self-attention** for cross-modal interaction with **semantic-aware expert routing** for task-specific FFN computation, reducing interference between language reasoning and motion planning.
- We design a flow-matching planner on the routed action-token pathway to refine noisy action tokens into future waypoints and temporal speed profiles under multimodal conditioning.
- Experiments on Bench2Drive show that **VECTOR-DRIVE** outperforms representative end-to-end and VLA-based baselines, with ablations validating expert routing, three-stage training, and flow-based action decoding.

II. Related Work

A. Vision-Language-Action Models for Autonomous Driving

Large-scale vision-language pretraining has promoted VLA models for autonomous driving. Existing methods typically adapt a pretrained VLM or LLM by jointly encoding visual observations, navigation cues, ego states, and language prompts, followed by a trajectory or control head [13], [15]–[17]. Recent works further improve driving-oriented semantic representations. FLARE learns future-aware latent representations from VLM features, while ReCogDrive combines VLM-based driving cognition with a diffusion planner for reasoning-guided planning [28], [29]. These studies demonstrate the value of VLM priors for long-tail scene understanding and interpretable driving decisions.

However, most VLA driving models still rely on largely shared computation for language reasoning and motion prediction. Although shared backbones preserve multimodal interaction, the same transformer and FFN parameters must support discrete language generation and continuous trajectory regression. Since these objectives differ in optimization behavior, fully shared computation may introduce task interference and weaken motion accuracy [22], [23]. Moreover, when semantic reasoning is connected to planning only through text, latent summaries, or high-level decisions, fine-grained trajectory generation is only indirectly conditioned on scene understanding, limiting reasoning-control coupling in interactive driving.

B. Decoupled Action Generation and Continuous Planning

Recent works decouple action generation from vision-language reasoning to improve continuous control. In robotics, π_0 introduces a flow-matching action model on top of a pretrained vision-language backbone [24]. This paradigm has also been explored in autonomous driving. DriveVLA-W0 introduces a lightweight action expert, while DriveMoE extends the π_0-style formulation with mixture-of-experts modules for scene-specialized perception and skill-specialized action generation [25], [26]. These designs reduce the burden on general-purpose VLM representations and strengthen action modeling.

Diffusion- and flow-based planners further improve continuous trajectory generation by modeling complex future distributions beyond deterministic regression [4], [24], [29]. However, fully separated reasoning-action pipelines often reduce the VLM to a conditioning module, weakening bidirectional interaction between semantic understanding and motion generation.

In contrast, our method keeps all modalities in a unified token sequence. Shared self-attention preserves cross-modal interaction at each layer, while token-type-aware FFN routing separates vision-language and trajectory-specific computation. Semantic reasoning and motion planning therefore remain tightly coupled through attention, while their FFN transformations are decoupled to reduce task interference.

III. Method

We formulate end-to-end driving as unified multimodal reasoning and motion generation. Built on Qwen2.5-VL-3B, the model maps visual observations, navigation conditions, language commands, ego states, and noisy action representations into an interleaved token sequence. A tightly coupled multimodal Transformer processes this sequence with shared self-attention and semantic-aware FFN routing. The LM Head generates chain-of-thought reasoning and driving instructions, while the Flow Head predicts future waypoints and temporal speed profiles. Fig. 2 illustrates the architecture of VECTOR-DRIVE.

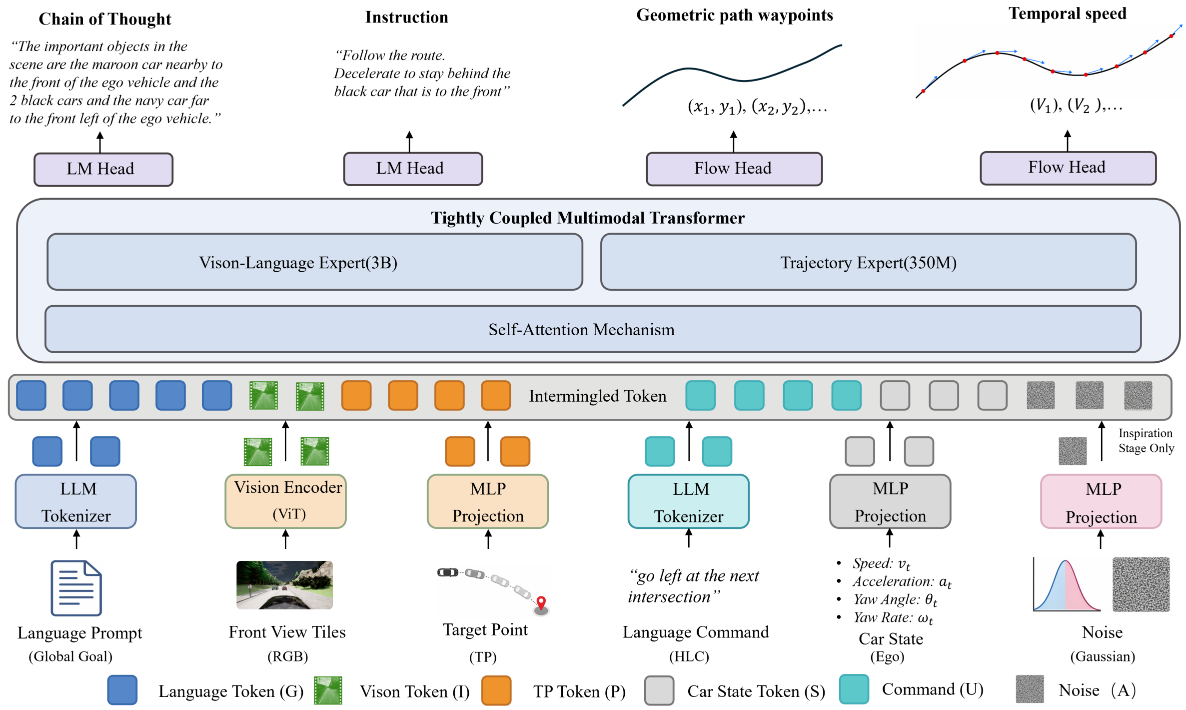

Fig. 2. **Overall architecture of VECTOR-DRIVE.** Visual observations, navigation conditions, language commands, ego states, and noisy action states are organized as an interleaved multimodal token sequence. Shared self-attention preserves cross-modal interaction, while semantic-aware FFN routing separates vision-language and trajectory-oriented computation.

A. Problem Formulation

We formulate end-to-end driving as conditional multimodal generation. Given a front-view RGB frame I, a global-goal prompt G, a target point P, a high-level command U, and an ego state S, the model jointly produces language-level reasoning and motion outputs. The ego state is

$$S = [v, a, \theta, \dot{\theta}], \tag{1}$$

where v, a, θ, and $\dot{\theta}$ denote ego speed, acceleration, yaw angle, and yaw rate.

The language output consists of chain-of-thought reasoning and driving instruction text:

$$O = \{o_1, \ldots, o_K\}. \tag{2}$$

The motion output contains a spatial path and a temporal speed profile:

$$Y^p = \{p_1, \ldots, p_{20}\}, \qquad p_k = (x_k, y_k) \in \mathbb{R}^2, \tag{3}$$

where one waypoint is predicted per meter over a 20-meter horizon, and

$$Y^v = \{v_1, \ldots, v_{10}\}, \qquad v_q \in \mathbb{R}, \tag{4}$$

where v_q is the speed at the q-th future step.

The conditional mapping is

$$(O, Y^p, Y^v) = \mathcal{F}(I, G, P, U, S), \tag{5}$$

where $\mathcal{F}$ denotes the proposed multimodal driving model. The language output describes scene-level reasoning and driving intent, while Y^p and Y^v provide the spatial path and temporal speed profile for downstream control.

B. Unified Tokenization

All modalities are mapped into a unified token space. The global-goal prompt G and high-level command U are tokenized into Z_G and Z_U. The front-view image I is encoded into visual tokens Z_I. The target point P and ego state $S = [v, a, \theta, \dot{\theta}]$ are projected into the shared hidden space by lightweight MLPs, producing Z_P and Z_S.

For flow-based motion generation, a noisy action representation is introduced. Let X_τ be the noisy path state at interpolation time τ, and let $e(\tau)$ be its time embedding. The noisy action tokens are

$$Z_A(\tau) = \phi(X_\tau, e(\tau)), \tag{6}$$

where $\phi(\cdot)$ denotes the action-token preprocessor.

All token groups are arranged into an interleaved multimodal sequence:

$$Z = \mathrm{Interleave}\left(Z_G, Z_I, Z_P, Z_U, Z_S, Z_A\right), \tag{7}$$

where the order follows Fig. 2, and z_i denotes the i-th token. This unified sequence enables cross-modal interaction before task-specific decoding.

C. Semantic-Aware Expert Routing

The interleaved sequence Z is used as the initial hidden sequence, $\mathbf{H}^{(0)} = Z$. Each Transformer layer contains shared self-attention and two expert-specific FFN branches. Routing is applied after attention: all tokens first exchange context through shared self-attention, and the contextualized states are then routed according to token semantics. This preserves semantic-to-motion interaction while separating task-specific feed-forward computation.

Given $\mathbf{H}^{(l-1)} \in \mathbb{R}^{N \times d}$ at layer $l \in \{1, \ldots, L\}$, shared self-attention is applied to the full sequence:

$$\tilde{\mathbf{H}}^{(l)} = \mathbf{H}^{(l-1)} + \mathrm{Attn}^{(l)}\left(\mathrm{LN}\left(\mathbf{H}^{(l-1)}\right)\right). \tag{8}$$

where $\mathrm{LN}(\cdot)$ denotes layer normalization. The attention operator uses shared Q, K, V, and O projections:

$$\mathrm{Attn}^{(l)}(\mathbf{X}) = \mathrm{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d}} + \mathbf{M}\right)\mathbf{V}\mathbf{W}_O^{(l)}. \tag{9}$$

where $\mathbf{Q} = \mathbf{X}\mathbf{W}_Q^{(l)}$, $\mathbf{K} = \mathbf{X}\mathbf{W}_K^{(l)}$, and $\mathbf{V} = \mathbf{X}\mathbf{W}_V^{(l)}$. Thus, action tokens aggregate semantic and driving-condition context at each layer.

After shared attention, let $\tilde{\mathbf{h}}_i^{(l)}$ denote the post-attention hidden state of the i-th token in $Z = \{z_1, \ldots, z_N\}$. We define $\mathcal{I}_{\mathrm{vl}} = \{i \mid z_i \in Z_G \cup Z_I \cup Z_U\}$ and $\mathcal{I}_{\mathrm{act}} = \{i \mid z_i \in Z_P \cup Z_S \cup Z_A\}$. The routed hidden sequences are obtained by ordered gathering:

$$\tilde{\mathbf{H}}_{\mathrm{vl}}^{(l)} = \mathcal{C}_{i\in\mathcal{I}_{\mathrm{vl}}}\left(\tilde{\mathbf{h}}_i^{(l)}\right), \tag{10}$$

$$\tilde{\mathbf{H}}_{\mathrm{act}}^{(l)} = \mathcal{C}_{i\in\mathcal{I}_{\mathrm{act}}}\left(\tilde{\mathbf{h}}_i^{(l)}\right). \tag{11}$$

where $\mathcal{C}(\cdot)$ denotes ordered concatenation along the token dimension following the original order in Z.

The routed expert inputs are

$$\mathbf{O}_1^{(l)} = \mathrm{LN}\left(\tilde{\mathbf{H}}_{\mathrm{vl}}^{(l)}\right), \qquad \mathbf{O}_2^{(l)} = \mathrm{LN}\left(\tilde{\mathbf{H}}_{\mathrm{act}}^{(l)}\right), \tag{12}$$

corresponding to the Vision-Language Expert and the Trajectory Expert.

Each expert uses a gated FFN with gate, up, and down projections:

$$\mathrm{FFN}_e^{(l)}(\mathbf{X}) = \left[\sigma\left(\mathbf{X}\mathbf{W}_{e,\mathrm{gate}}^{(l)}\right) \odot \left(\mathbf{X}\mathbf{W}_{e,\mathrm{up}}^{(l)}\right)\right]\mathbf{W}_{e,\mathrm{down}}^{(l)}, \tag{13}$$

where e indexes the expert branch, with $e = 0$ for the Vision-Language Expert and $e = 1$ for the Trajectory Expert. Here, $\sigma(\cdot)$ is SiLU, and $\odot$ denotes element-wise multiplication.

The expert outputs are

$$\mathbf{O}_{1y}^{(l)} = \mathrm{FFN}_0^{(l)}(\mathbf{O}_1^{(l)}), \qquad \mathbf{O}_{2y}^{(l)} = \mathrm{FFN}_1^{(l)}(\mathbf{O}_2^{(l)}). \tag{14}$$

They are then merged back into the original token order:

$$\mathbf{H}^{(l)} = \mathrm{Merge}_Z\left(\tilde{\mathbf{H}}_{\mathrm{vl}}^{(l)} + \mathbf{O}_{1y}^{(l)},\ \tilde{\mathbf{H}}_{\mathrm{act}}^{(l)} + \mathbf{O}_{2y}^{(l)}\right). \tag{15}$$

where $\mathrm{Merge}_Z(\cdot)$ restores the interleaved order.

This attention–routing–merge process repeats at every layer. After merging, motion tokens remain conditioned on refined semantic features, while the causal mask prevents action-to-language information leakage. Thus, shared attention conditions motion generation on semantics, while separate FFNs reduce task interference. The routing is static and token-type-aware rather than learned top-k routing, avoiding additional router parameters and load-balancing losses.

To improve waypoint-level consistency while preventing action leakage into language representations, we use a causal multimodal attention mask and allow bidirectional attention only within noisy action tokens:

$$M_{ij} = \begin{cases} 0, & z_i, z_j \in Z_A, \\ M_{\mathrm{causal}}(i,j), & \text{otherwise.} \end{cases} \tag{16}$$

With Z_A placed at the end of the sequence, action tokens can attend to preceding semantic and driving-condition tokens, but language-side tokens cannot access future action states.

D. Generation of Chain-of-Thought and Driving Instruction

The language branch generates chain-of-thought reasoning and driving instruction text from the final routed Vision-Language hidden states $\mathbf{H}_{\mathrm{vl}}^{(L)}$. The LM Head performs autoregressive decoding:

$$p(O \mid \mathbf{H}_{\mathrm{vl}}^{(L)}) = \prod_{k=1}^{K} p(o_k \mid o_{<k}, \mathbf{H}_{\mathrm{vl}}^{(L)}). \tag{17}$$

Language decoding is performed only from Vision-Language hidden states, while alignment with motion planning is encouraged by the shared multimodal context and joint optimization. Under the causal mask, action tokens attend to semantic and driving-condition tokens, enabling semantic-conditioned trajectory generation without future-action leakage into language decoding.

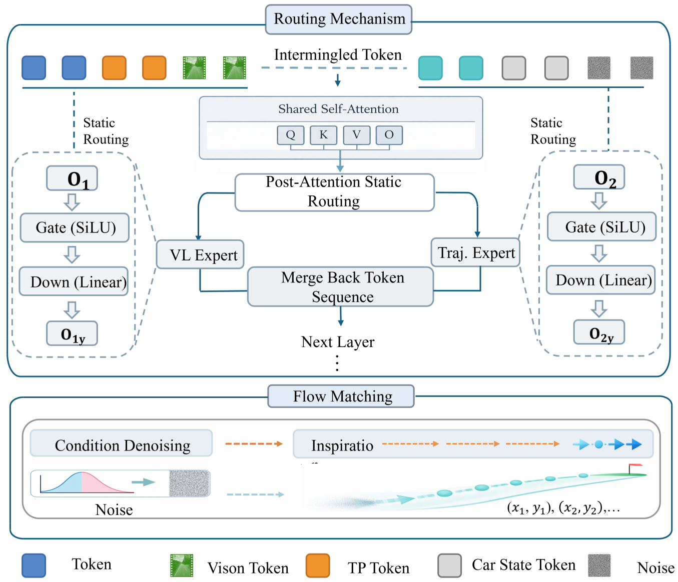

Fig. 3. **Routing and planning. Top:** Interleaved tokens are processed by shared self-attention, routed to the Vision-Language or Trajectory Expert, and merged back into the sequence. **Bottom:** The flow planner refines Gaussian noise into future trajectories under semantic and driving-condition guidance.

E. Flow-Matching Action Planner with Multimodal Conditioning

Fine-grained motion is generated by a flow-matching planner for both spatial path prediction and temporal speed profiling. The spatial branch models future route geometry, and the speed branch captures temporal motion evolution. We detail the spatial formulation and apply the same flow-matching principle to speed generation.

Spatial path generation. The spatial branch predicts a 20-meter future path with 20 ordered waypoints. Let $Y^p \in \mathbb{R}^{20\times 2}$ denote the ground-truth path, and let $\boldsymbol{\epsilon} \sim \mathcal{N}(0, I)$ be Gaussian noise with the same shape. For $\tau \in [0,1]$, the noisy path state is

$$X_\tau = (1-\tau)\boldsymbol{\epsilon} + \tau Y^p, \tag{18}$$

with the target vector field

$$\mathbf{v}^\star = Y^p - \boldsymbol{\epsilon}. \tag{19}$$

At each τ, X_τ and its time embedding $e(\tau)$ are encoded into noisy action tokens $Z_A(\tau)$ by Eq. (6), which replace Z_A at the corresponding action-token positions in the interleaved sequence. After shared attention, expert routing, and merge-back operations, the final action-token hidden states are obtained by ordered gathering:

$$\mathbf{H}_A^{(L)}(\tau) = \mathcal{C}_{i\in\mathcal{I}_A}\left(\mathbf{h}_i^{(L)}(\tau)\right), \qquad \mathcal{I}_A = \{i \mid z_i \in Z_A\}. \tag{20}$$

Thus, $Z_A(\tau)$ provides the current noisy state, while $\mathbf{H}_A^{(L)}(\tau)$ provides the semantic-conditioned motion representation used by the Flow Head.

The implicit semantic condition is obtained from the final Vision-Language hidden states:

$$\mathbf{C}_{\mathrm{imp}}^{(L)} = \psi_{\mathrm{vl}}\left(\mathbf{H}_{\mathrm{vl}}^{(L)}\right), \tag{21}$$

where $\psi_{\mathrm{vl}}(\cdot)$ projects semantic hidden states into the flow-head input space. The explicit driving condition is formed by projecting ego state, navigation representation, and target point:

$$\mathbf{C}_{\mathrm{exp}} = \mathcal{C}\left(\psi_S(S), \psi_{\mathrm{nav}}(\bar{z}_{\mathrm{nav}}), \psi_P(P)\right), \tag{22}$$

where

$$\bar{z}_{\mathrm{nav}} = \mathrm{Pool}\left(\mathcal{C}(Z_G, Z_U)\right). \tag{23}$$

The complete flow condition is

$$\mathbf{C}_{\mathrm{fm}}^{(L)} = \mathcal{C}\left(\mathbf{C}_{\mathrm{imp}}^{(L)}, \mathbf{C}_{\mathrm{exp}}\right). \tag{24}$$

The Flow Head predicts the vector field from the routed action hidden states and the multimodal condition:

$$\mathbf{F}_\tau = \mathcal{C}\left(\psi_A\left(\mathbf{H}_A^{(L)}(\tau)\right), \mathbf{C}_{\mathrm{fm}}^{(L)}\right), \tag{25}$$

$$\hat{\mathbf{v}}_\omega\left(X_\tau, \tau \mid \mathbf{C}_{\mathrm{fm}}^{(L)}\right) = f_\omega\left(\mathbf{F}_\tau\right), \tag{26}$$

where $f_\omega(\cdot)$ denotes the flow head.

At inference, the path is initialized from Gaussian noise:

$$X_0 \sim \mathcal{N}(0, I), \tag{27}$$

and the conditional ODE

$$\frac{dX_\tau}{d\tau} = \hat{\mathbf{v}}_\omega\left(X_\tau, \tau \mid \mathbf{C}_{\mathrm{fm}}^{(L)}\right), \qquad \tau \in [0,1], \tag{28}$$

is solved to obtain $\hat{Y}^p$. During ODE solving, the current state X_τ is repeatedly converted into $Z_A(\tau)$, processed by the routed Transformer, and decoded by the Flow Head.

Temporal speed generation. The model also predicts a future speed profile:

$$\hat{Y}^v = \{\hat{v}_1, \ldots, \hat{v}_{10}\}, \qquad \hat{v}_q \in \mathbb{R}. \tag{29}$$

Speed generation follows the same flow-matching formulation as spatial path generation. It uses the same implicit semantic condition $\mathbf{C}_{\mathrm{imp}}^{(L)}$ and explicit driving condition $\mathbf{C}_{\mathrm{exp}}$, while replacing the noisy path state with a noisy speed-profile state. The corresponding routed speed hidden states are then decoded by a speed-specific Flow Head.

Thus, spatial path and temporal speed generation share a unified multimodal flow-matching scheme, with separate heads for route geometry and temporal motion evolution.

F. Three-Stage Training Strategy

We train the framework in three stages to stabilize vision-language reasoning, specialize motion generation, and align both pathways.

Stage I: Vision-Language Learning. The Trajectory Expert and motion heads are frozen, while the Vision-Language Expert and LM Head are optimized for scene understanding, semantic reasoning, and instruction generation:

$$\mathcal{L}_{\mathrm{lang}} = -\sum_{k=1}^{K} \log p\left(o_k \mid o_{<k}, \mathbf{H}_{\mathrm{vl}}^{(L)}\right), \tag{30}$$

where $\mathbf{H}_{\mathrm{vl}}^{(L)}$ denotes the final routed Vision-Language hidden states.

Stage II: Driving-Skill Learning. The Vision-Language Expert is frozen, while the Trajectory Expert, flow heads, and condition projectors are optimized for motion generation. For the spatial path branch, the flow-matching loss is

$$\mathcal{L}_{\mathrm{path}} = \mathbb{E}_{Y^p,\boldsymbol{\epsilon}^p,\tau}\left[\left\| f_\omega^p\left(\mathbf{F}_\tau^p\right) - \left(Y^p - \boldsymbol{\epsilon}^p\right)\right\|_2^2\right], \tag{31}$$

where $\mathbf{F}_\tau^p$ is formed by concatenating the projected routed path-token hidden states with $\mathbf{C}_{\mathrm{fm}}^{(L)}$.

We further use a second-order finite-difference regularizer:

$$\mathcal{L}_{\mathrm{smooth}} = \frac{1}{18}\sum_{k=2}^{19} \left\|\hat{p}_{k+1} - 2\hat{p}_k + \hat{p}_{k-1}\right\|_2^2. \tag{32}$$

The speed branch follows the same flow-matching formulation:

$$\mathcal{L}_{\mathrm{speed}} = \mathbb{E}_{Y^v,\boldsymbol{\epsilon}^v,\tau}\left[\left\| f_\eta^v\left(\mathbf{F}_\tau^v\right) - \left(Y^v - \boldsymbol{\epsilon}^v\right)\right\|_2^2\right], \tag{33}$$

where $\mathbf{F}_\tau^v$ is constructed analogously by replacing routed path-token hidden states with routed speed-profile hidden states.

The driving objective is

$$\mathcal{L}_{\mathrm{drive}} = \lambda_{\mathrm{path}}\mathcal{L}_{\mathrm{path}} + \lambda_{\mathrm{smooth}}\mathcal{L}_{\mathrm{smooth}} + \lambda_{\mathrm{speed}}\mathcal{L}_{\mathrm{speed}}, \tag{34}$$

where λ_{path}, $\lambda_{\mathrm{smooth}}$, and λ_{speed} are balancing coefficients.

Stage III: Joint Full-Parameter Fine-Tuning. All experts and prediction heads are unfrozen and optimized end-to-end:

$$\mathcal{L} = \mathcal{L}_{\mathrm{lang}} + \mathcal{L}_{\mathrm{drive}}. \tag{35}$$

This schedule first adapts the language pathway, then specializes the trajectory pathway, and finally aligns semantic reasoning with flow-based motion generation under shared multimodal context.

IV. Experiments

We evaluate VECTOR-Drive on closed-loop performance, qualitative behavior, and component-wise ablations. The experiments compare VECTOR-Drive with representative end-to-end and VLA-based baselines on Bench2Drive and analyze semantic-aware routing, three-stage training, and flow-based trajectory decoding.

A. Experimental Settings

Closed-loop driving benchmark. We evaluate on Bench2Drive, a CARLA-based closed-loop benchmark with 44 interactive scenarios and 5 routes per scenario, yielding 220 routes under diverse traffic and weather conditions. We report Driving Score (DS), Success Rate (SR), Efficiency, Comfort, and Multi-Ability. Multi-Ability measures fine-grained behaviors including merging, overtaking, braking, give-way, and traffic-sign compliance.

Implementation details. The model is built upon Qwen2.5-VL-3B. Both the Vision-Language Expert and Trajectory Expert are initialized from the corresponding pretrained FFN layers of the backbone; the latter is then specialized for

TABLE I
MAIN RESULTS AND MULTI-ABILITY ON BENCH2DRIVE [12]. HIGHER IS BETTER FOR ALL METRICS.

Method	Closed-loop metrics ↑				Multi-Ability (%) ↑					
	DS	SR (%)	Efficiency	Comfort	Merging	Overtake	Brake	Give-Way	Traffic-Sign	Mean
TCP [7]	40.70	15.00	54.26	47.80	16.18	20.00	20.00	10.00	6.99	14.63
TCP-ctrl [7]	30.47	7.27	55.97	**51.51**	10.29	4.44	10.00	10.00	6.45	8.23
TCP-traj [7]	59.90	30.00	76.54	18.08	8.89	24.29	51.67	40.00	46.28	34.22
ThinkTwice [8]	62.44	31.23	69.33	16.22	27.38	18.42	35.82	**50.00**	54.23	37.17
DriveAdapter [9]	64.22	33.08	70.22	16.01	28.82	26.38	48.76	**50.00**	56.43	42.08
AD-MLP [30]	18.05	0.00	48.45	22.63	0.00	0.00	0.00	0.00	4.35	0.87
UniAD-Tiny [2]	40.73	13.18	123.92	47.04	8.89	9.33	20.00	20.00	15.43	14.73
UniAD-Base [2]	45.81	16.36	129.21	43.58	14.10	17.78	21.67	10.00	14.21	15.55
VAD [10]	42.35	15.00	157.94	46.01	8.11	24.44	18.64	20.00	19.15	18.07
DriveTransformer [11]	63.46	35.01	100.64	20.78	17.57	35.00	48.36	40.00	52.10	38.60
ReCogDrive [29]	71.36	45.45	138.18	17.45	29.73	20.00	69.09	20.00	71.34	42.03
DriveMoE [26]	74.22	48.64	175.96	15.31	34.67	40.00	65.45	40.00	59.44	47.91
Orion [18]	77.74	54.62	151.48	17.38	25.00	71.11	78.33	30.00	69.15	54.72
AutoVLA [19]	78.84	57.73	146.93	39.33	–	–	–	–	–	–
SimLingo [17]	85.07	67.27	**259.23**	33.67	53.75	68.89	81.67	**50.00**	82.11	67.28
VECTOR-DRIVE (Ours)	**88.91**	**71.82**	252.10	34.20	**57.50**	**78.33**	**91.67**	**50.00**	**82.63**	**72.03**

motion-oriented computation. The training set is self-collected in CARLA using the privileged rule-based expert PDM-lite, and contains approximately 3 million frames covering diverse traffic interactions, routes, and environmental conditions. Training uses 4×A800 80GB GPUs with a learning rate of 3×10^{-5}. The three stages use 10, 12, and 7 epochs, respectively. Loss weights are set to $\lambda_{\text{path}} = 1.0$, $\lambda_{\text{smooth}} = 0.1$, and $\lambda_{\text{speed}} = 1.0$.

For the flow-matching planner, the future path is initialized from Gaussian noise and refined with an Euler ODE solver. During inference, 10 Euler steps produce the final 20-waypoint trajectory, and the speed branch predicts a 10-step future speed profile.

B. Main Closed-loop Driving Results

As summarized in Table I, VECTOR-DRIVE achieves 88.91 DS and 71.82% SR on Bench2Drive, outperforming SimLingo by 3.84 DS and 4.55 percentage points in SR. It also improves over ReCogDrive and DriveMoE, representing decoupled and weakly coupled VLA paradigms, respectively. These results indicate that shared attention with expert-specific FFN routing provides an effective interface for transferring scene-level semantics into executable motion plans.

VECTOR-DRIVE further achieves the highest Multi-Ability mean score of 72.03%, with clear gains in merging, overtaking, braking, and traffic-sign compliance. While SimLingo shows slightly higher Efficiency, VECTOR-DRIVE performs better in DS, SR, and behavior-level robustness, suggesting improved closed-loop reliability in interactive driving scenarios.

TABLE II
ABLATION ON THE COUPLING AND ROUTING DESIGN. DRIVING SCORE (DS) AND SUCCESS RATE (SR) ARE REPORTED. HIGHER IS BETTER.

Method	DS ↑	SR (%) ↑
Decoupled VLM	83.33	64.09
Weakly Coupled VLM	86.67	68.18
Learned Routing	88.85	71.82
Single Action-Oriented Expert	87.67	70.45
VECTOR-DRIVE	**88.91**	**71.82**

C. Qualitative Driving Visualization

The qualitative cases in Fig. 4 further illustrate the temporal consistency of the generated decisions. In the car-following scenario, VECTOR-DRIVE maintains stable motion when the surrounding traffic is sparse, decelerates as lateral space becomes constrained by nearby vehicles, and resumes acceleration after the interaction is resolved. In the stop-controlled right-turn scenario, the model first yields at the stop line, then performs a cautious creeping behavior to observe cross traffic, and finally accelerates once a safe gap is available. These behaviors suggest that the generated language responses and speed profiles are temporally aligned with the closed-loop driving context.

D. Ablation Studies

To isolate the effect of semantic-motion coupling and expert routing, Table II compares several architectural variants. Decoupled VLM removes the action expert and uses VLM representations to condition motion generation, obtaining 83.33 DS and 64.09% SR. Weakly Coupled VLM further decouples the shared-attention interaction between vision-language and action-related tokens, keeping only a coarse semantic conditioning interface; it improves to 86.67 DS and 68.18% SR

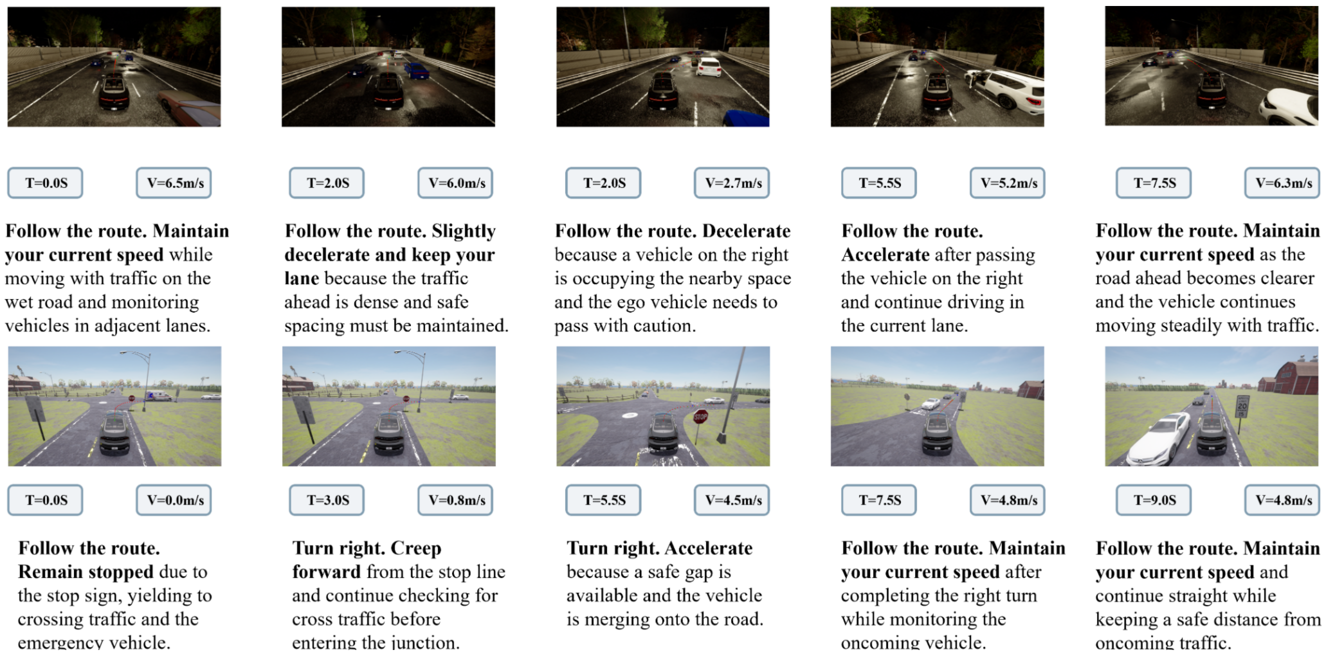

Fig. 4. **Qualitative closed-loop visualization.** Two scenarios are shown with time stamps, speed, and language-guided responses. **Top:** In nighttime wet-road car following, the model maintains speed, decelerates for dense traffic and a nearby right-side vehicle, and accelerates after the interaction resolves. **Bottom:** At a stop-controlled right turn, it stops, creeps forward to check cross traffic, accelerates after a safe gap, and proceeds steadily.

TABLE III
ABLATION ON THE THREE-STAGE TRAINING STRATEGY. DRIVING SCORE (DS) AND SUCCESS RATE (SR) ARE REPORTED. HIGHER IS BETTER.

Method	DS ↑	SR (%) ↑
Joint training only	85.67	69.09
w/o Stage I VL learning	87.57	70.91
w/o Stage II driving learning	88.47	71.36
w/o Stage III joint fine-tuning	88.59	71.68
Full three-stage training	**88.91**	**71.82**

TABLE IV
ABLATION ON THE MOTION DECODING HEAD. DRIVING SCORE (DS) AND SUCCESS RATE (SR) ARE REPORTED. HIGHER IS BETTER.

Method	DS ↑	SR (%) ↑
VLM+Regression	86.01	67.27
VLM+DDPM	87.67	70.45
VECTOR-DRIVE	**88.91**	**71.82**

but still lags behind the full model. These results suggest that robust planning requires both motion-specialized computation and token-level semantic-motion interaction. Single Action-Oriented Expert also reduces performance, highlighting the importance of preserving a dedicated Vision-Language pathway during motion learning.

Learned Routing reaches 88.85 DS and 71.82% SR, close to the proposed static routing strategy. This indicates that the main gain comes from shared-attention coupling and expert-specific FFN computation, rather than from a more complex learned router. We therefore adopt static semantic-aware routing to avoid additional router parameters and load-balancing losses.

Table III evaluates the three-stage training strategy. Direct joint training reaches 85.67 DS and 69.09% SR, indicating that jointly optimizing language reasoning and motion prediction from the start is less stable. Removing Stage I leads to a larger drop than removing Stage II or Stage III, suggesting that early vision-language adaptation provides the key semantic priors for motion learning, while the later stages mainly offer additional refinement.

Finally, Table IV studies the choice of motion decoder under the same backbone. The regression decoder obtains 86.01 DS and 67.27% SR, while the DDPM decoder improves the result to 87.67 DS and 70.45% SR. The proposed flow-matching planner achieves 88.91 DS and 71.82% SR, demonstrating that continuous vector-field decoding is more effective for refining noisy action tokens into executable trajectories under multimodal conditioning.

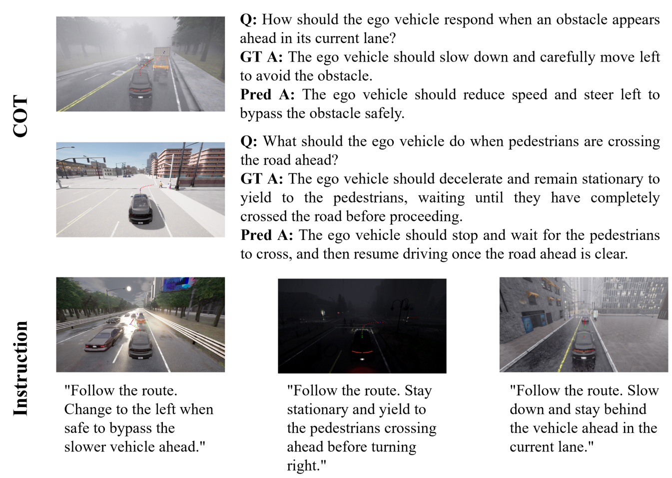

Fig. 5. **CoT and instruction visualization.** The examples show scene-aware reasoning and concise driving instructions generated by VECTOR-DRIVE under different traffic conditions.

E. CoT and Instruction Visualization

Fig. 5 visualizes the language-generation ability of VECTOR-DRIVE. Although language quality is not evaluated as a standalone metric, the examples show that the model produces scene-aware reasoning and concise instructions consistent with the intended driving behaviors. Together with the Stage-I ablation in Table III, this suggests that language-side semantic learning provides useful contextual priors for closed-loop planning.

F. Discussion

The results indicate that effective VLA driving requires both semantic-motion coupling and task-specific computation. VECTOR-DRIVE achieves stronger performance in interaction-heavy scenarios, including merging, overtaking, braking, and traffic-sign compliance, suggesting that semantic cues are better converted into executable motion plans. The ablations show that weakening this coupling, removing the Vision-Language pathway, or replacing the flow planner degrades closed-loop performance. Overall, shared attention preserves cross-modal interaction, while expert-specific FFNs reduce interference between language reasoning and trajectory generation.

V. CONCLUSION

We presented VECTOR-DRIVE, a tightly coupled VLA framework for end-to-end autonomous driving. By combining shared self-attention, semantic-aware FFN routing, and flow-based action decoding, the model jointly supports scene reasoning and continuous motion generation. Experiments on Bench2Drive demonstrate strong closed-loop performance, while ablations verify the benefits of expert routing, progressive training, and flow-matching planning. These results show that coupling multimodal interaction at the attention level while separating task-specific FFN computation is effective for robust autonomous driving.

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